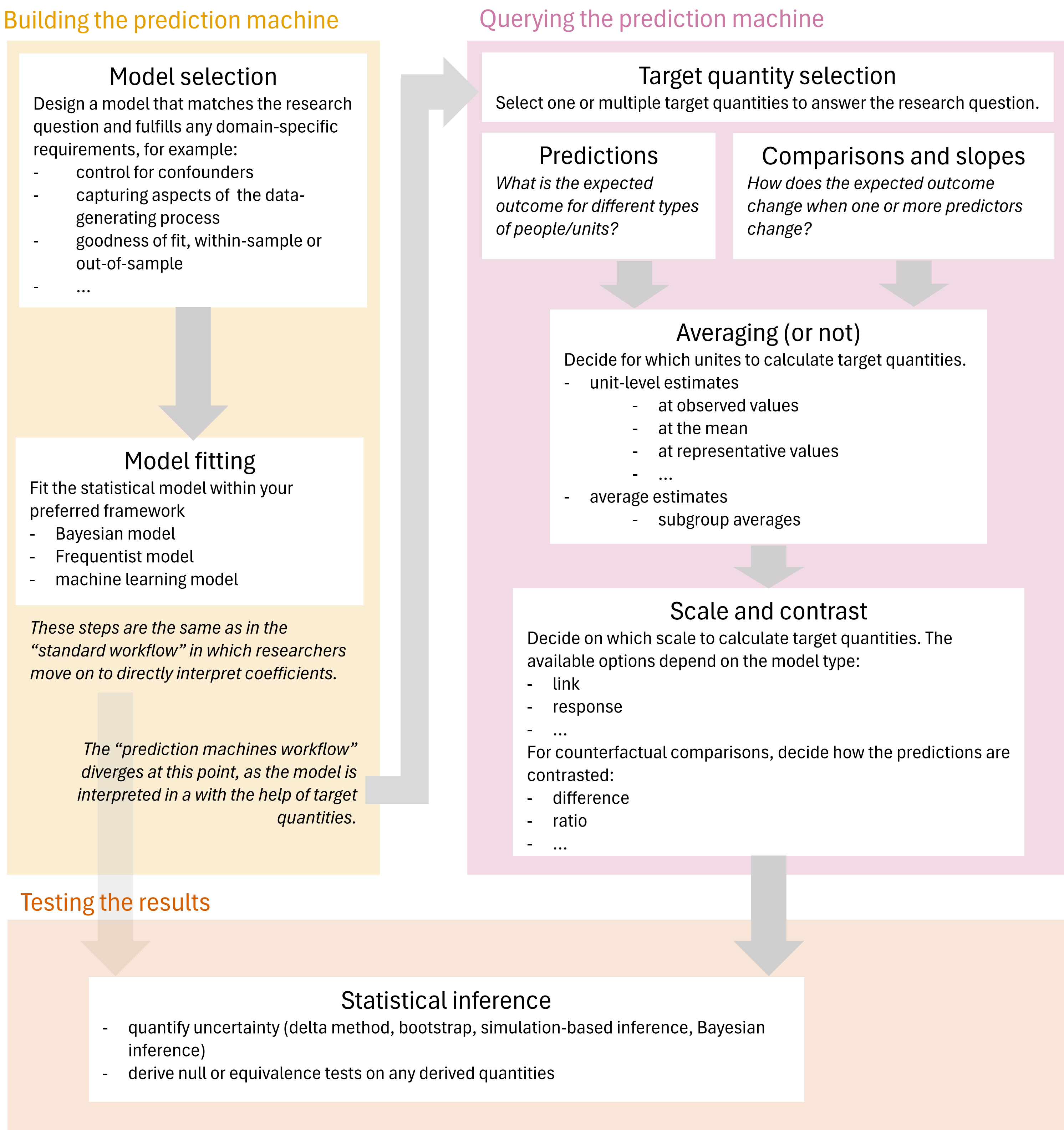
Building the prediction machine
Model selection
Design a model that matches the research question and fulfills any domain-specific requirements, for example:
- control for confounders
- capturing aspects of the data-generating process
- goodness of fit, within-sample or out-of-sample
- ...
Model fitting
Fit the statistical model within your preferred framework
- Bayesian model
- Frequentist model
- machine learning model
These steps are the same as in the “standard workflow” in which researchers move on to directly interpret coefficients.
The “prediction machines workflow” diverges at this point, as the model is interpreted in a with the help of target quantities.
Querying the prediction machine
Target quantity selection
Select one or multiple target quantities to answer the research question.
Predictions
What is the expected outcome for different types of people/units?
Comparisons and slopes
How does the expected outcome change when one or more predictors change?
Averaging (or not)
Decide for which unites to calculate target quantities.
- unit-level estimates
- at observed values
- at the mean
- at representative values
- ...
- average estimates
- subgroup averages
Scale and contrast
Decide on which scale to calculate target quantities. The available options depend on the model type:
- link
- response
- ...
For counterfactual comparisons, decide how the predictions are contrasted:
- difference
- ratio
- ...
Testing the results
Statistical inference
- quantify uncertainty (delta method, bootstrap, simulation-based inference, Bayesian inference)
- derive null or equivalence tests on any derived quantities